A learned collision operator for coarse-cell DSMC

ES-Fokker–Planck with learned rates and cubic drift, trained through the solver

2026-11-13 (intermediate state)

1. The operator: ES-FP + reconstruction + learned cubic drift

In each coarse collision cell, every step:

- reconstructed target $u(x_i) = u_c + G_c(x_i - \bar{x}_c)$, peculiar $c_i = v_i - u(x_i)$; $\Pi = \langle cc^T \rangle$, $\theta = \frac{1}{3}\operatorname{tr}\Pi$
- rates $\quad r_s = e^{\delta_s}\, p/\mu(T), \quad r_q = e^{\delta_q}\,\mathrm{Pr}\, p/\mu(T)$, $\mathrm{Pr} = \frac{2}{3}$
- $a = e^{-r_q \Delta t/3}, \quad e_s = e^{-r_s \Delta t}$ (raised where needed: Mathiaud–Mieussens ν-clamp)

$$c_i' = a\, c_i + \mathrm{chol}(\Sigma)\, \xi_i + (1-a^3)\, \beta \Big(\tfrac{|c_i|^2}{\theta} - 5 \Big) c_i,$$

$$\Sigma = \theta(1-a^2) I + (e_s - a^2)\, \Pi_{\mathrm{dev}}, \qquad \xi_i \sim \mathcal{N}(0, I)$$

then a projection restores each cell's momentum and energy exactly. Exactly integrated: stress relaxes at r_s, heat flux at r_q, for any Δt; the cubic term adds $\approx 27(1-a^3)\beta\, q$ to the heat flux, ≈ 0 to stress.

Learned per cell (bounded, zero $\Rightarrow$ the physical operator):

output	acts on	physics
$\delta_s \in (-2, 2)$	$\log r_s$	viscosity
$\delta_q \in (-2, 2)$	$\log r_q$	conductivity
$\beta \in (-\frac{1}{2}, \frac{1}{2})$	cubic drift	heat flux via $\langle \lvert c \rvert^4 c \rangle$

$(\delta_s, \delta_q, \beta) = B \tanh\big(f_\theta(\phi_c/\sigma)/B\big)$, ϕ_c: 9 cell features.

Why: coarse cells relax every particle toward one cell state $\Rightarrow$ excess transport $\propto (\Delta x/\lambda)^2$ (NTC $+23\%$, ES-FP+recon $+11\%$ viscosity at $\Delta x = 2\lambda$). **Potential:** $\Delta x = \lambda/4 \to 2\lambda$ is $8\times$ per axis $= 64\times$ fewer particles in 2-D ($512\times$ in 3-D) at fixed particles/cell; a pathwise-differentiable, conservative, Galilean/rotation-invariant operator.

2. The features: one term per known error mechanism

All local, dimensionless, invariant (λ, τ the cell's own VSS mean free path and time); each standardised by its RMS over the training cells.

feature	form	mechanism
dx2	$(\Delta x/\lambda)^2$	cell-size error of uniform in-cell collisions, $\mu_{\rm eff}/\mu - 1 = \frac{16}{45\pi}(\Delta x/\lambda)^2$ (Alexander–Garcia–Alder)
dx4	$(\Delta x/\lambda)^4$	next order of the same expansion
dt2	$(\Delta t/\tau)^2$	time-step error
dt1	$\Delta t/\tau$	residual first-order splitting error
inv_n	$1/N_c$	finite-particle bias of the cell state
dx2_inv_n	$(\Delta x/\lambda)^2/N_c$	cell-size error measured with a noisy cell state
kn_u2	$(\lambda\lvert S\rvert/v_{\rm th})^2$	Burnett-order (Chapman–Enskog $O(\mathrm{Kn}^2)$) stress closure
kn_t2	$(\lambda\lvert\nabla \ln T\rvert)^2$	Burnett-order heat-flux closure
dx2_kn2	$(\Delta x/\lambda)^2(\mathrm{Kn}_u^2 + \mathrm{Kn}_T^2)$	cell-size error growing with gradients (the reconstruction is linear)

Not used: the 1-D collective (particles-per-λ) term – non-local, suppressed in 2-D slabs; a positive-definiteness-limiter term – tried and dropped (it slows stress relaxation only $\sim$2% in the M10 core, and is mostly sampling noise).

3. Learning through the solver

Data (no gradient): fine NTC at $\Delta x = \lambda/4$, 20 particles/cell – 4 shear waves ($u_0/v_{\mathrm{th}} \in \{0.1, 0.5\}$, $L \in \{32, 128\}\lambda$) and 4 normal shocks (M 1.5, 3, 6, 10); coarse cases at $\Delta x = 2\lambda$ (waves), λ_1 and $4\lambda_1$ (shocks). Train S3 S4 K1 K4 K8, test the other 7.

Window: two independent coarse replicas, 5 blocks $\times$ 20 steps; waves from thinned fine snapshots, shocks from *persistent chains* at the model's own steady state (snapshot starts hid the heat-flux deficit: $\int q_x$ 1.01 in the window vs 0.94 steady).

Deviation: $d_{tkf} = P\big[F(m_t) - F(m_t^{\mathrm{ref}})\big]_{kf}$, F: P_{xy} (waves, Fourier mode), τ_{xx}, q_x per cell (shocks, reference placed at each replica's shock); $w_f = 1/\langle F_{\mathrm{ref}}^2 \rangle$.

Objective $\frac{1}{2} w\, \mathbb{E}[d]^2$; running-bias estimator ($b \approx \mathbb{E}[d]$, stop-gradient):

$$\nabla_\theta J = \operatorname*{mean}_{t,k} \sum_f w_f\, b_{tkf}\, \tfrac{1}{2} \left(\tfrac{\partial m^a}{\partial \theta} + \tfrac{\partial m^b}{\partial \theta} \right)_{tkf}$$

- $\partial m/\partial\theta$: **exact reverse-mode AD** through the 100 steps (checkpointed scan, keys replayed) – collision (rates, Cholesky, cubic, projection, features), free flight, inflow; discrete events carry no gradient
- cost $\approx$ a few forward windows; no fine-grid adjoint, no ensemble terms
- cases summed, Adam with fan-in-normalised steps, tail-averaged iterates

Checks: AD = FD per window (common random numbers); estimator sign vs. steady FD at the base. Open: the window start is held fixed (truncation of the steady-state response).

4. The linear model and the Lasso: which mechanisms matter

Model: $(\delta_s, \delta_q, \beta) = A\,\phi_c/\sigma, \quad A \in \mathbb{R}^{3\times 9}$ (18 coefficients without the cubic row). Each coefficient = the weight of one error mechanism on one transport coefficient: directly interpretable.

Gauss–Newton Lasso: per window $d(A) \approx d(A_0) + J(A - A_0)$, J by forward-mode AD (one tangent per coefficient), averaged per case:

$$\min_{A} \ \tfrac{1}{2} \sum_{\text{cases}} \operatorname*{mean}_{\text{rows}} w\big(\bar{d} + \bar{J}(A - A_0)\big)^2 + \alpha \|A\|_1$$

coordinate descent along the α path; α by leave-one-case-out CV (one-s.e. rule); stability selection over 200 window bootstraps; optional sign constraints (cell-size/time-step terms may only speed up stress relaxation).

First result (snapshot windows, before chains):

- kept 2 of 18, stability 0.99–1.0: $1/N_c$ and $(\Delta x/\lambda)^2 \mathrm{Kn}^2$, both speeding up relaxation (lower viscosity/conductivity); no Prandtl-row term
- free and sign-constrained fits identical
- squared bias $4.5 \times 10^{-3} \to 3.0 \times 10^{-3}$
- held-out (LOCO) error improves for S3, S4, K4, K8 at moderate α, not K1; small α overfits (held-out K4 error $\times 15$)

To redo with chains and the cubic row; the linear fits also serve as the MLP's interpretable baseline (linear $\approx$ MLP $\Rightarrow$ the theory basis suffices).

5. The MLP and its cross-validated design

Model: $f_\theta(\phi) = \mathrm{MLP}(\phi/\sigma) - \mathrm{MLP}(0)$, tanh hidden layers, zero-initialised last layer $\Rightarrow$ starts as the physical operator and vanishes in resolved equilibrium cells; same bounded outputs. Depth 2, width 32 $\approx$ 1.4k parameters.

Design sweep (L9 orthogonal array, every pair of levels once):

factor	levels
hidden layers	1, 2, 3
width	8, 32, 128
weight decay	0, 10^{-3}, 10^{-2}
learning rate	10^{-3}, 3×10^{-3}, 10^{-2}

Per design point, 7 runs: 5 leave-one-case-out folds (train on 4 of S3 S4 K1 K4 K8, validate forward on the held-out one) and 2 all-case runs with different seeds (training variance; seeds also pick each run's shock chains from the shared warm pool). 63 runs, $\sim$215 iterations each, sized to 55 GPU-h on 7 MIG slices (one shared warm pool, built once).

Selection: held-out error of the folds; then the chosen design trained on all five and evaluated on the 7 test cases and a coarse M10 cylinder.

Steps: Adam divided by each layer's fan-in (lr = output change per iteration, independent of width); decoupled weight decay; mean of the last 25% of iterates.

Status (2026-11-13)

- **Base operators** (coarse, untrained): ES-FP+recon removes about half of NTC's cell-size viscosity error; its shock stress excess grows with Mach ($\int \tau_{xx}$ 1.05 at M1.5 $\rightarrow$ 1.17 at M10), the heat flux is 3–13% low.
- **Robust so far**: shear-wave viscosity corrected in every run (S3 1.065 $\rightarrow$ 0.97–1.05, S4 1.131 $\rightarrow$ 1.04–1.08); M10 stress partly (1.17 $\rightarrow$ 1.04–1.16).
- **Diagnosed**: (i) snapshot-started windows never saw the steady heat-flux deficit $\Rightarrow$ persistent chains; (ii) chain pairs compared against one target at their mean shock position $\Rightarrow$ anti-correlated replicas (negative pair loss), a smeared-shock fake bias, and K4 worsening under its own gradient – fixed (per-replica targets, bias table in the shock frame).
- **Running**: re-checks with the fix (K4/K8 alone, linear/MLP, two rates vs. cubic), then the MLP sweep.
- **Open**: window truncation (the start state is held fixed); what two rates can express in non-Gaussian shock cores (the cubic row); the weak M1.5 signal.